

Online Faculty Development Program

On

“Decoding AI: An AIML Workshop Experience”



Session Details : Day 6
Topic : Discussion on ML Deployment and Production

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Machine Learning: A Comprehensive Overview

Join us on a journey to explore the fascinating world of Machine Learning. From its historical roots to its modern applications, we'll unpack the key concepts, techniques, and potential of this transformative technology.

History of Machine Learning

The roots of Machine Learning can be traced back to the mid-20th century, where researchers were interested in creating machines capable of learning from data.

Early pioneers like Alan Turing and Arthur Samuel laid the groundwork for the field, exploring concepts like artificial intelligence and game-playing algorithms.

The development of computers and the availability of large datasets fueled rapid advancements in the field. Key milestones included the invention of the perceptron, a simple artificial neuron, and the development of statistical learning methods.

What is Machine Learning?

1 Learning from Data

At its core, Machine Learning is about enabling computers to learn from data without being explicitly programmed. Instead of relying on hard-coded rules, ML algorithms identify patterns and make predictions based on the data they are trained on.

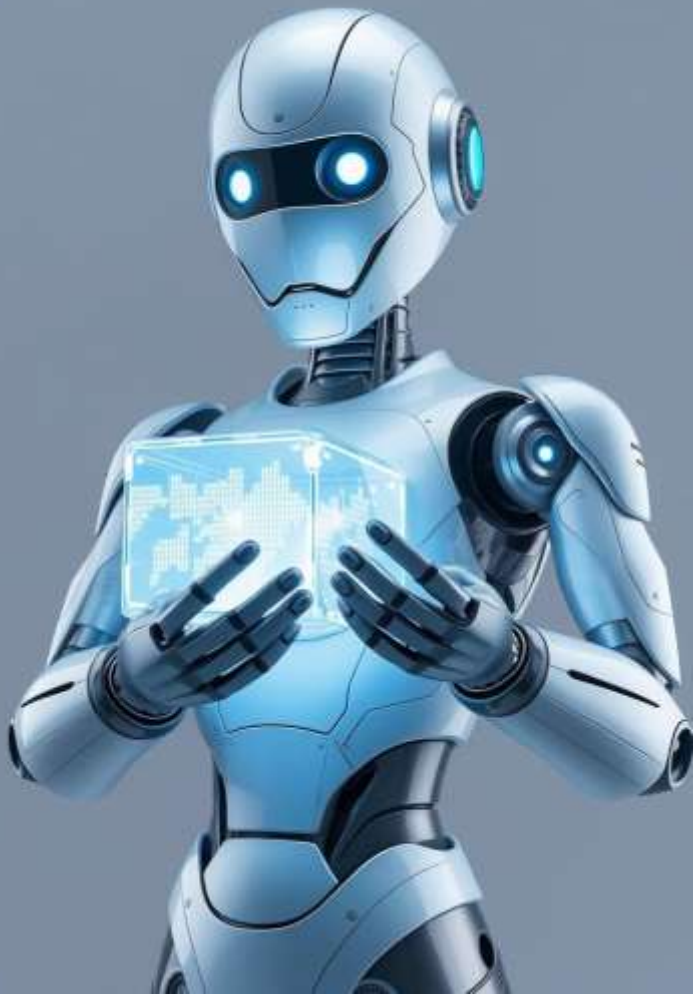
3 Continuous Improvement

The power of Machine Learning lies in its ability to continuously improve its performance based on new data. As it encounters more information, it refines its understanding of the underlying patterns and relationships.

2 Algorithms and Models

Machine Learning uses algorithms to create models that can predict outcomes or make decisions. These models are developed by training them on large datasets, allowing them to learn and adapt over time.





Why Machine

Automation

Machine Learning automates tasks that are repetitive, complex, or require human intuition, freeing up time and resources for more strategic endeavors.

Personalization

ML algorithms power personalized experiences across various industries, from recommending products and content to tailoring healthcare treatments and financial services.

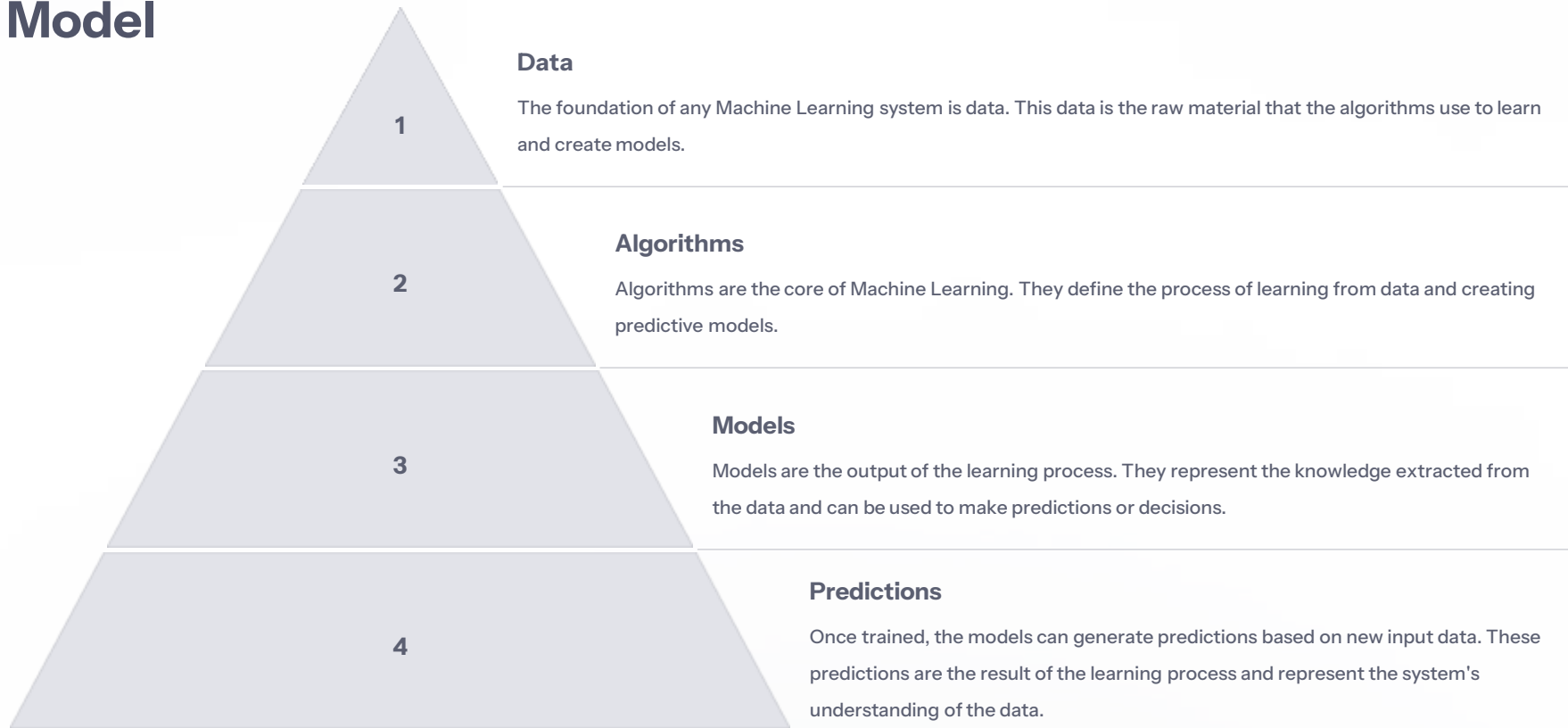
Insights and Predictions

Machine Learning helps us understand data in deeper ways, uncovering hidden patterns and making accurate predictions about future trends and outcomes.

Innovation

Machine Learning is driving innovation across all sectors, creating new products, services, and solutions that enhance efficiency, productivity, and quality of life.

Learning System Model



Training and Testing

1

Training Data

The training phase involves feeding the algorithm a large set of labeled data, allowing it to learn the underlying patterns and relationships.

2

Model Building

Based on the training data, the algorithm builds a model that captures the key insights from the data and represents the relationships learned.

3

Testing Data

The testing phase involves using a separate set of data that the model has not seen before to evaluate its performance and accuracy.



Fredases



Tacilevices



Performance Evaluation

1

Accuracy

Measures how often the model correctly predicts the outcome.

3

Recall

Measures how many of the actual positive cases were correctly identified by the model.

2

Precision

Measures how many of the positive predictions made by the model were actually correct.

4

F1 Score

A balance between precision and recall, providing a comprehensive measure of model performance.

Machine Learning Algorithms



Decision Trees

Hierarchical structures that use a series of decisions to classify data points.



Neural Networks

Inspired by the human brain, these networks consist of interconnected nodes that learn from data.



Linear Regression

Used to predict a continuous outcome based on a linear relationship between variables.



Clustering

A technique that groups similar data points together based on their characteristics.

Machine Learning Structure

Supervised Learning

Models are trained on labeled data, where the desired outcome is known, allowing the algorithm to learn the relationship between inputs and outputs.

Unsupervised Learning

Models are trained on unlabeled data, where the algorithm identifies patterns and relationships without explicit guidance.

Reinforcement Learning

Models learn through trial and error, receiving rewards for desired actions and penalties for undesirable ones.

Importance of Model Deployment

Turning Insights into Action

Deployment bridges the gap between theoretical models and practical applications, enabling businesses to leverage the predictive power of ML for informed decision-making.

Unlocking Value

Once deployed, ML models can automate tasks, optimize processes, and deliver personalized experiences, ultimately generating business value.



Deployment Transforms Theoretical Models into Practical Tools

1

Training

Models are trained on historical data to learn patterns and relationships.

2

Deployment

Models are packaged and deployed into a production environment where they can receive real-time data and make predictions.

3

Action

Predictions from deployed models inform decision-making, leading to improved outcomes.

Key Deployment Strategies



On-Premises

Deploy models on dedicated hardware within your own data center.



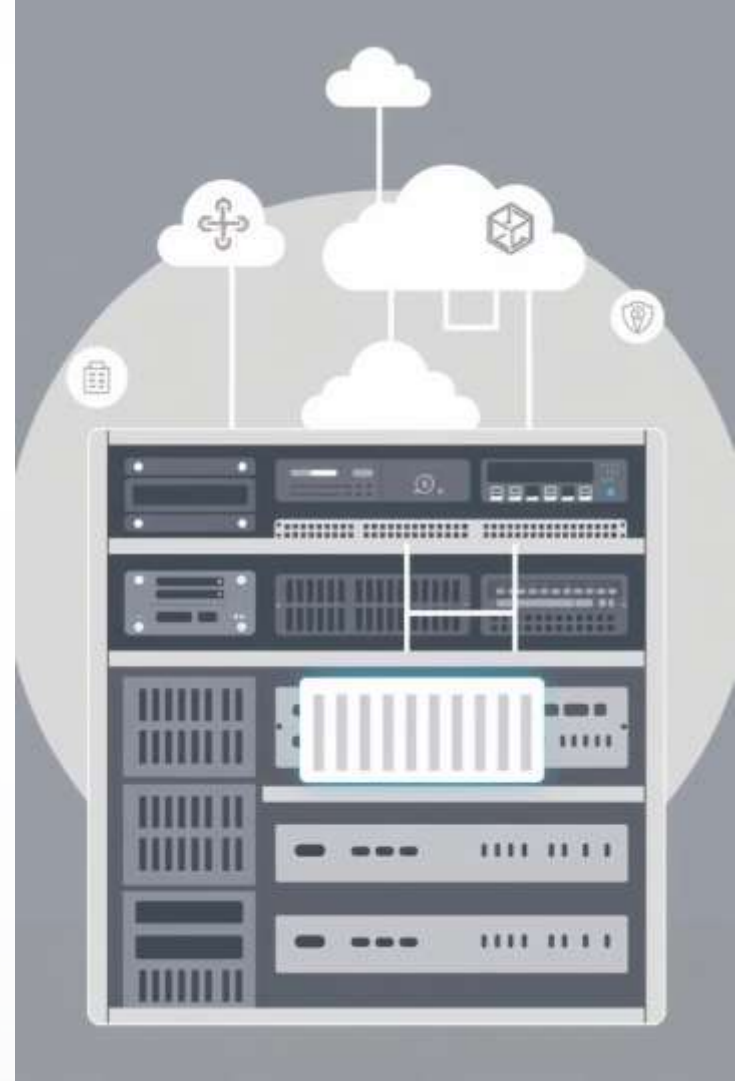
Containerization

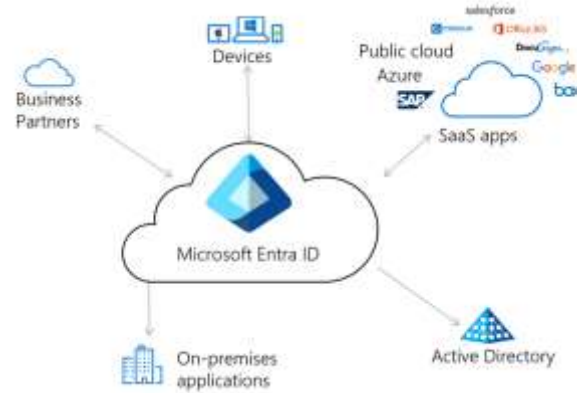
Package models in containers for portability and consistent execution across different environments.



Cloud-Based

Leverage cloud services to host and manage models.





On-Premises Deployment

Control

Provides complete control over hardware and software.

Security

Ensures data security and compliance for sensitive applications.

Cost

Can be more cost-effective for high-volume workloads.



Cloud-Based Deployment

Scalability

Easily scale resources up or down based on demand.

Cost-Efficiency

Pay-as-you-go pricing models can reduce infrastructure costs.

Managed Services

Cloud providers offer managed services for deployment and maintenance.

Containerization for Portability



Docker

Popular containerization technology that packages models and dependencies into a single, portable unit.



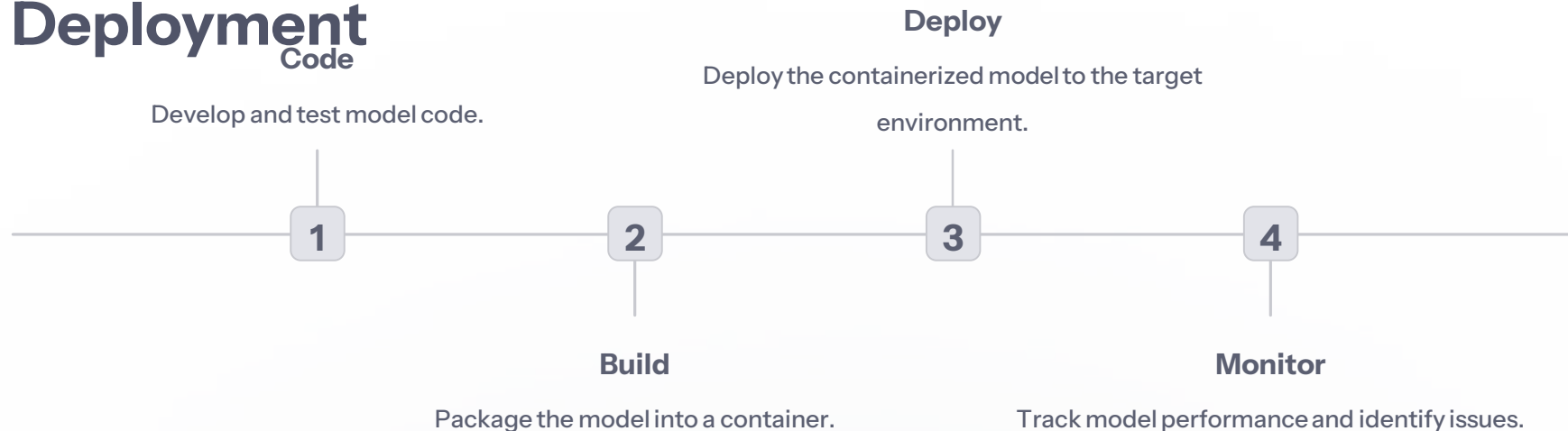
Kubernetes

Orchestration platform for managing and deploying containerized models at scale.

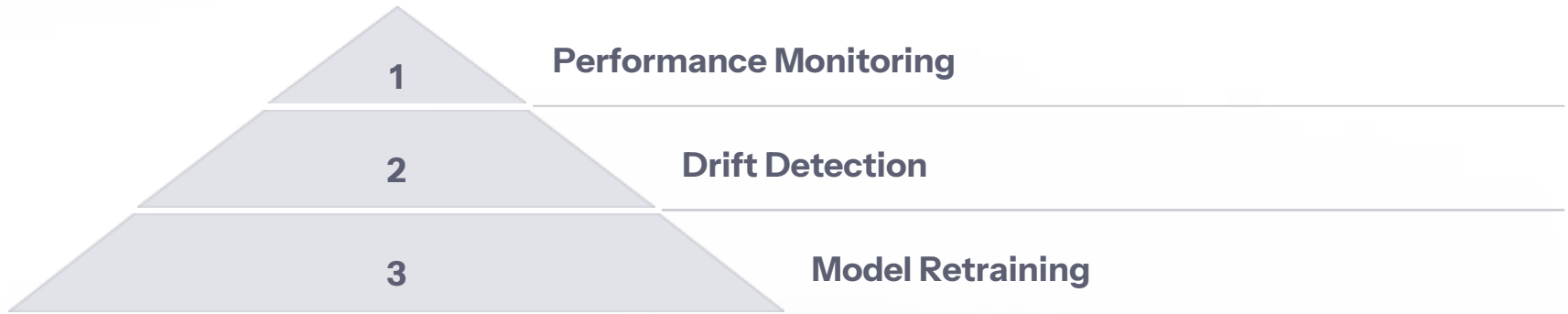




Continuous Integration and Deployment



Monitoring and Updating Deployed Models





Challenges in ML Model Deployment

1

Data Quality

Deploying models on real-world data can expose inconsistencies and quality issues that were not present in the training data.

2

Scalability

Models need to be able to handle increasing data volumes and user requests.

3

Security

Protecting sensitive data and model integrity is paramount.

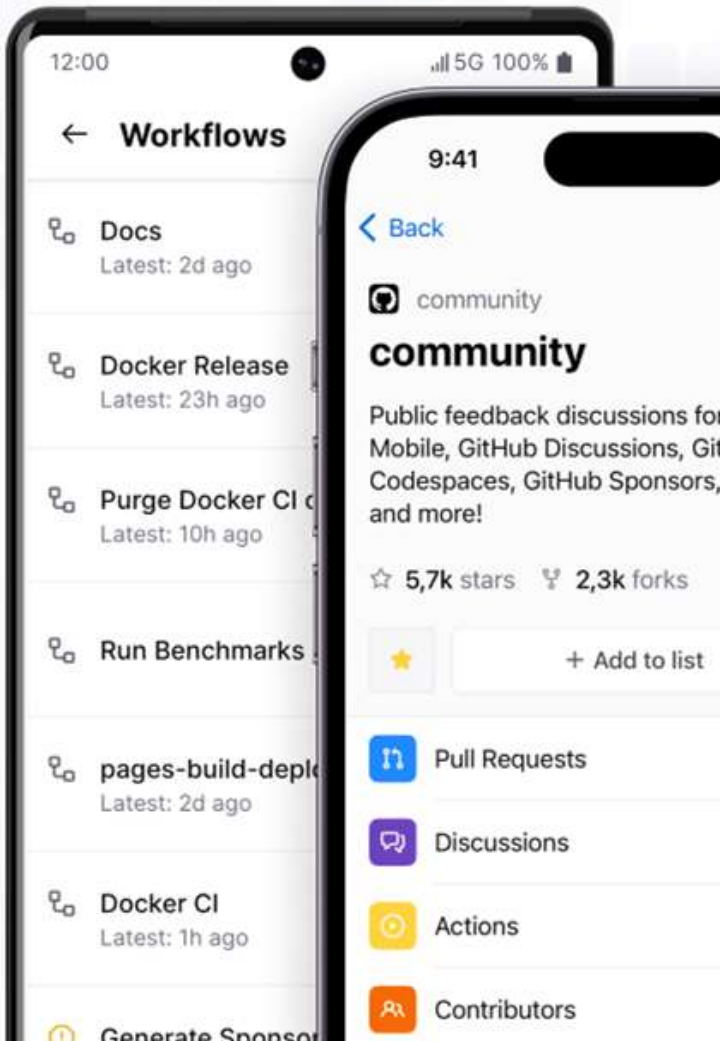
What is Heroku?

Cloud Platform

Heroku is a cloud platform as a service (PaaS) that enables developers to build, run, and operate applications entirely in the cloud.

Streamlit Integration

Heroku seamlessly integrates with Streamlit, allowing you to deploy your machine learning web apps with ease.



Creating a GitHub Repository

Create Repo

Start by creating a new GitHub repository for your Streamlit application.

Add Files

Add your Streamlit code, dependencies, and any other necessary files to this repository.

README

A README file is essential to document your project, providing instructions and context for others.

Navigating GitHub



GitHub Account

Log in to your GitHub account.



New Repository

Click on the "New repository" button.





Adding Files to Your Repository

1

Main Branch

Navigate to the main branch of your repository.

2

Add File

Click on the "Add File" button.

3

Create New File

Choose "Create New File" from the dropdown menu.

Creating a New File

1

Dropdown Options

Select "Create New File" from the dropdown options.

2

File Name

Enter a name for your new file.

3

Add Content

Paste your Streamlit code into the file.



Heroku Dashboard

1

Heroku Dashboard

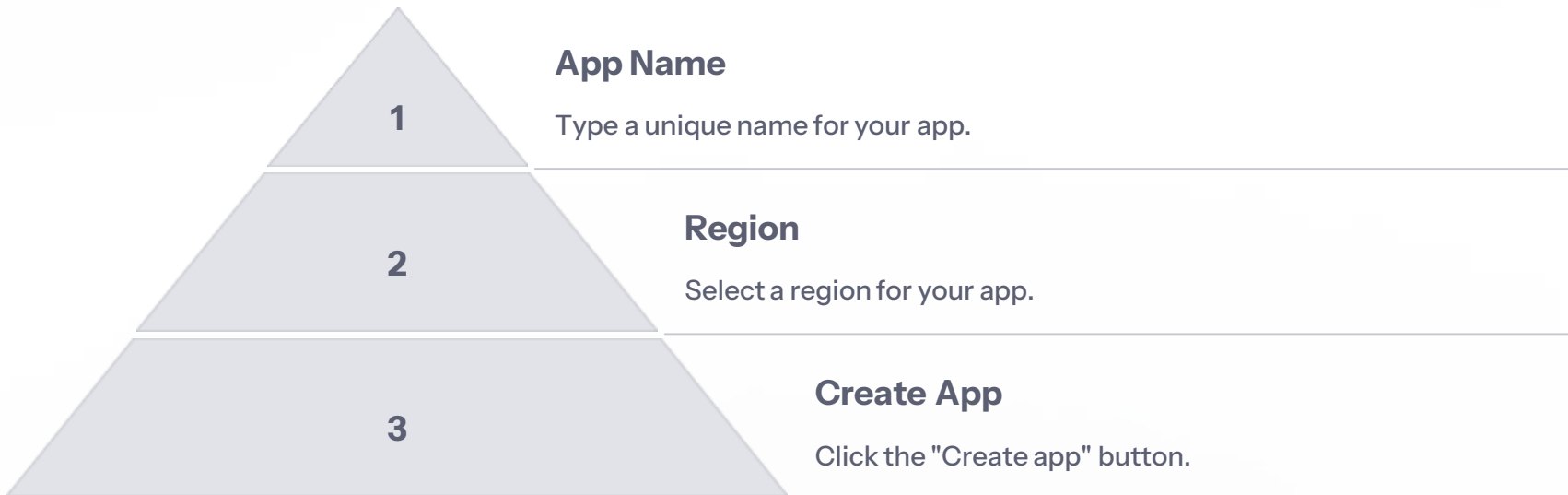
Visit your Heroku dashboard.

2

Create App

Click on the "Create new app" button.

Naming Your App



Heroku CLI Installation

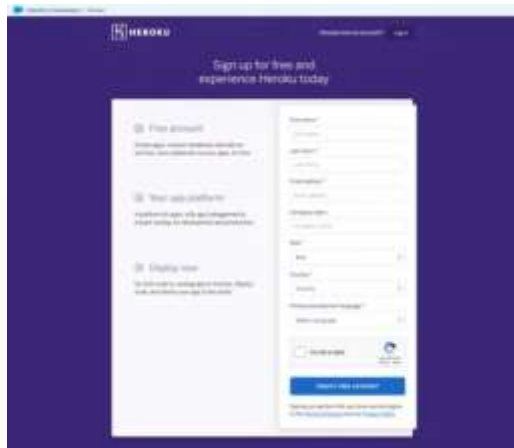
Heroku CLI

The Heroku CLI is a command-line interface for managing your Heroku applications.

Installation

Install the Heroku CLI on your local machine by following the instructions on the Heroku website.

Logging into Heroku



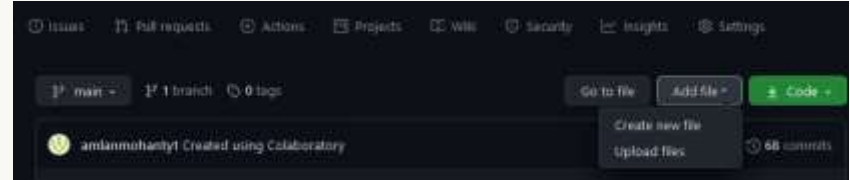
Use the Heroku CLI to log into your Heroku account and start deploying your Streamlit app.

Deploying a Streamlined Web App to Heroku

This presentation will guide you through the process of deploying a Streamlit web app to Heroku, a cloud platform for building and running applications.



Step 2: Create a GitHub Repository



Procfile

The Procfile contains the code that tells Heroku which files to execute when the application is opened. Add the line "web: gunicorn app:app" to your Procfile.

requirements.txt

The requirements.txt file lists the packages and dependencies needed for running the web app. For example: "streamlit" "gunicorn" "pandas" "numpy" "scikit-learn".

setup.sh

The setup.sh file contains shell script required to set up the shell environment. Copy the code from the image to your setup.sh file.

Step 3: Create a New Heroku App

1 Create New App

Visit your Heroku dashboard and click on "Create new app".

2 Name Your App

Type the name you want to give to your app. A green tick will appear beside your app name if the name is available. Then click on "Create app".

3 Open Your App

Your app will be created and you can view it by clicking on "Open app" in the top right corner of your page.



Step 5: Connect Your App to GitHub



Connect to GitHub

Go back to your Heroku page and connect your app to your GitHub repository. From the "Deployment method", click on "Connect to GitHub" or simply on the GitHub icon.



Enable Automatic Deploys

Click on "Enable Automatic Deploys".



Search for Your Repository

Select your GitHub account and search for your repository name. Your repository name will appear automatically after you click on the "Search" button. Click on "Connect".



Step 6: Start the Build Progress



Initial Release

Once you have completed all the previous steps, you can notice that your app's initial release has already started and Logplex is enabled from the "Activity" section or the "Overview" section.

Trigger the Build

To start the "Build Progress" so that your app is finally deployed, you have to follow a little trick. Go back to your GitHub repository and make any little change so that the build can finally start.

[Edit README.md](#)

I would suggest editing the README.md file and making any unnoticeable and irrelevant change. After you edit your repo and commit changes, the process of "build progress" begins.



Step 7: Wait for Your App to Deploy

Deployment

Everything is done on your part by now. Just sit back, relax and wait for your app to get deployed. It will take 2-5 mins to complete the process.

1

2

3

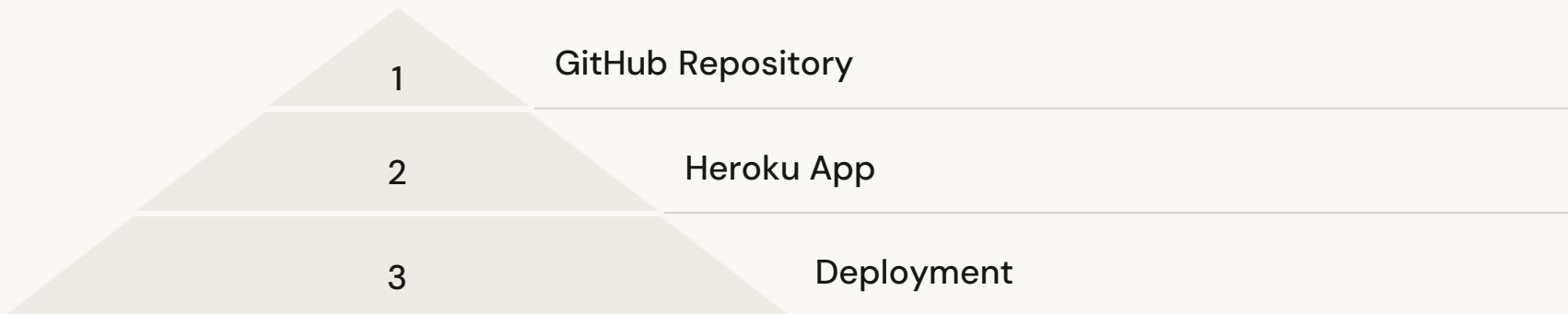
App Deployed

You will get a message saying that your app has been deployed to Heroku. Simply click on "Open app" in the top right corner or copy the app link from the "Build Log" to view your app.

View Build Progress

Rather than waiting around, go to the "Activity" or "Overview" section and click on "View build progress" to understand what is happening when the build is in progress.

Key Takeaways



This presentation has shown you how to deploy a Streamlined web app to Heroku. By following these steps, you can easily share your web app with others and make it accessible online.

Next Steps

- 1 Customize Your App
- 2 Add More Features
- 3 Deploy to Other Platforms

Now that you have successfully deployed your Streamlit web app to Heroku, you can explore further customization options. You can add more features, enhance the user interface, and even deploy your app to other platforms like AWS or Google Cloud.

ML Deployment projects:

1. Deploy your Machine Learning web app (Streamlit) on Heroku
2. Deploy a Machine Learning Model using Streamlit Library
3. Deploy Machine Learning Model using Flask
4. Deploying ML Models as API using FastAPI



Conclusion

Deploying ML models into production environments is a multi-faceted process requiring careful planning, robust infrastructure, and continuous monitoring. By following best practices and leveraging the right tools, organizations can ensure their models provide reliable and valuable insights in real-world applications. Effective deployment transforms theoretical models into actionable tools that can drive business decisions and generate significant value.